

Predicting credit card default

Comparing KNN, Logistic Regression and Random Forest with recall as the metric that matters

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30,000

clients (UCI dataset)

3

tuned classifiers

SDG 8

Decent Work & Economic Growth

Why credit default matters for SDG 8

1 The problem

Can we predict whether a credit card client will fail to pay next month?

Banks need to spot risky clients early. A good model supports credit-risk management and responsible lending, helping both institutions and customers avoid financial trouble.

**SDG
8**

Decent Work and Economic Growth

Target 8.10

Strengthen financial institutions and expand access to financial services.

Target 8.3

Links economic development with access to financial services.

A missed default is the costly mistake

User: credit-risk analyst at a bank or card provider. **Affected:** the credit card clients.

FALSE NEGATIVE

Predicted safe, but the client defaults

The client's credit risk is underestimated and the institution absorbs the loss. This is the mistake we most want to avoid.

FALSE POSITIVE

Predicted default, but the client pays

The client is treated as riskier than they are, which can trigger unnecessary risk-management actions.

So our main metric is recall on the default class: of all real defaulters, how many do we catch?

The data: 30,000 clients in Taiwan

30,000

rows

23

predictor features

22.1%

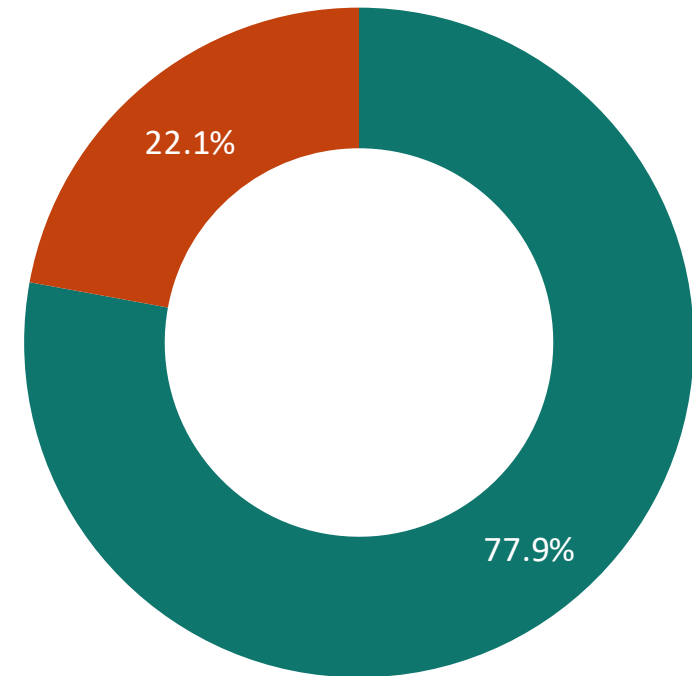
default (6,636 clients)

0

missing values

Source: UCI “Default of Credit Card Clients” (Yeh, 2009), CC BY 4.0. Data covers April–September 2005 and includes credit limit, demographics, repayment history, bill amounts and past payments.

Class balance: imbalanced target



■ No default ■ Default

How we built it

1

Load

Download the UCI dataset and drop the ID column.

2

Split

80/20 stratified train/test split (random_state=42).

3

Preprocess

StandardScaler on numbers, OneHotEncoder on categories, inside a Pipeline.

4

Tune

5-fold cross-validation, scored on recall.

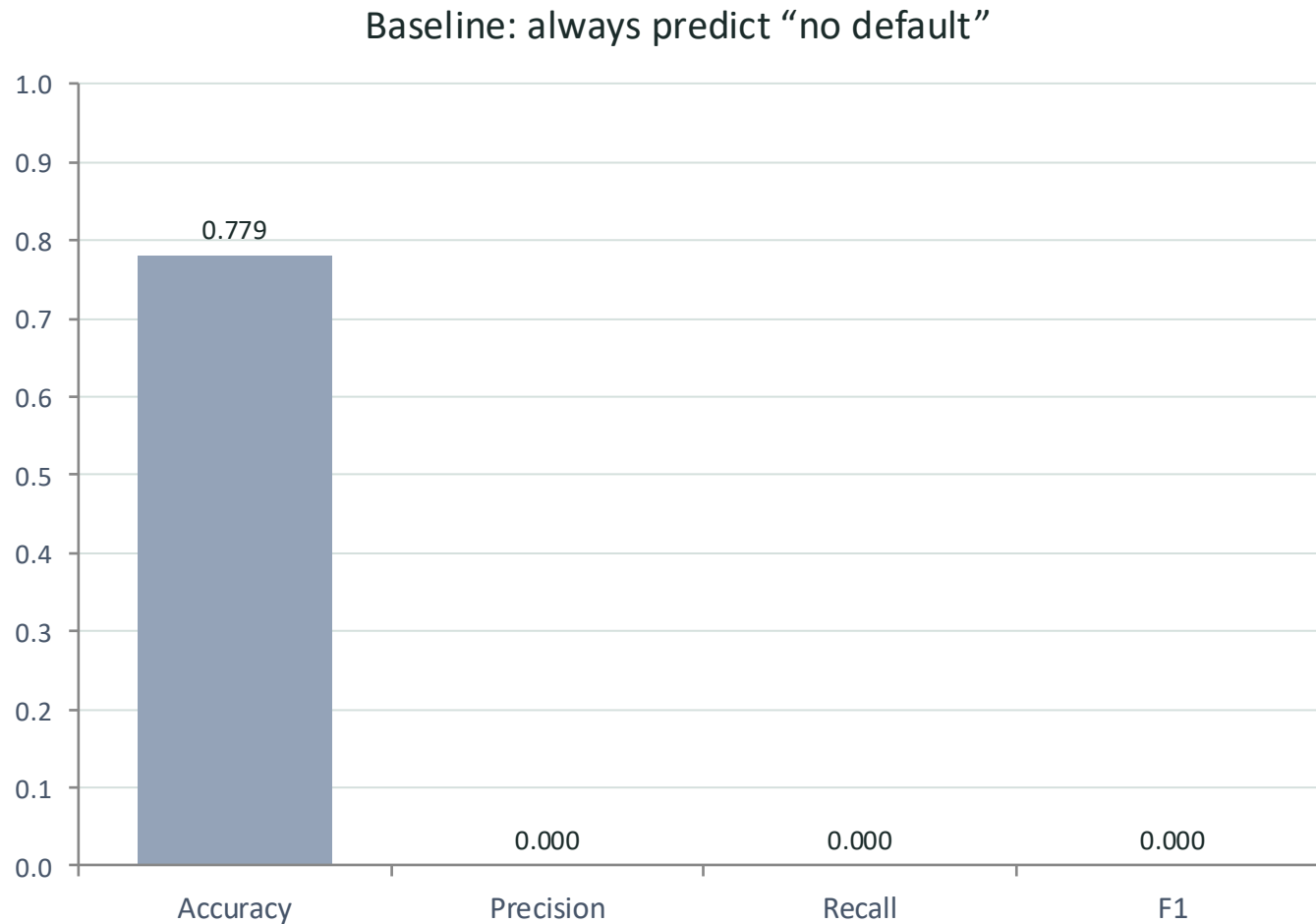
5

Evaluate

Compare on the held-out test set against a baseline.

Splitting first and keeping preprocessing inside the Pipeline means the test data never influences training, which avoids data leakage.

Why accuracy alone is misleading



77.9%

accuracy, from a model that never flags a single defaulter

0.000

recall on the default class. This is why recall is our primary metric.

Three models, each tuned for recall

K-Nearest Neighbours

Classifies by the closest training examples.

BEST PARAMETERS

n_neighbors = 3
weights = distance

0.355 ± 0.009

5-fold CV recall

Logistic Regression

A linear model that outputs default probabilities.

BEST PARAMETERS

C = 10

0.360 ± 0.012

5-fold CV recall

Random Forest

Many decision trees voting together.

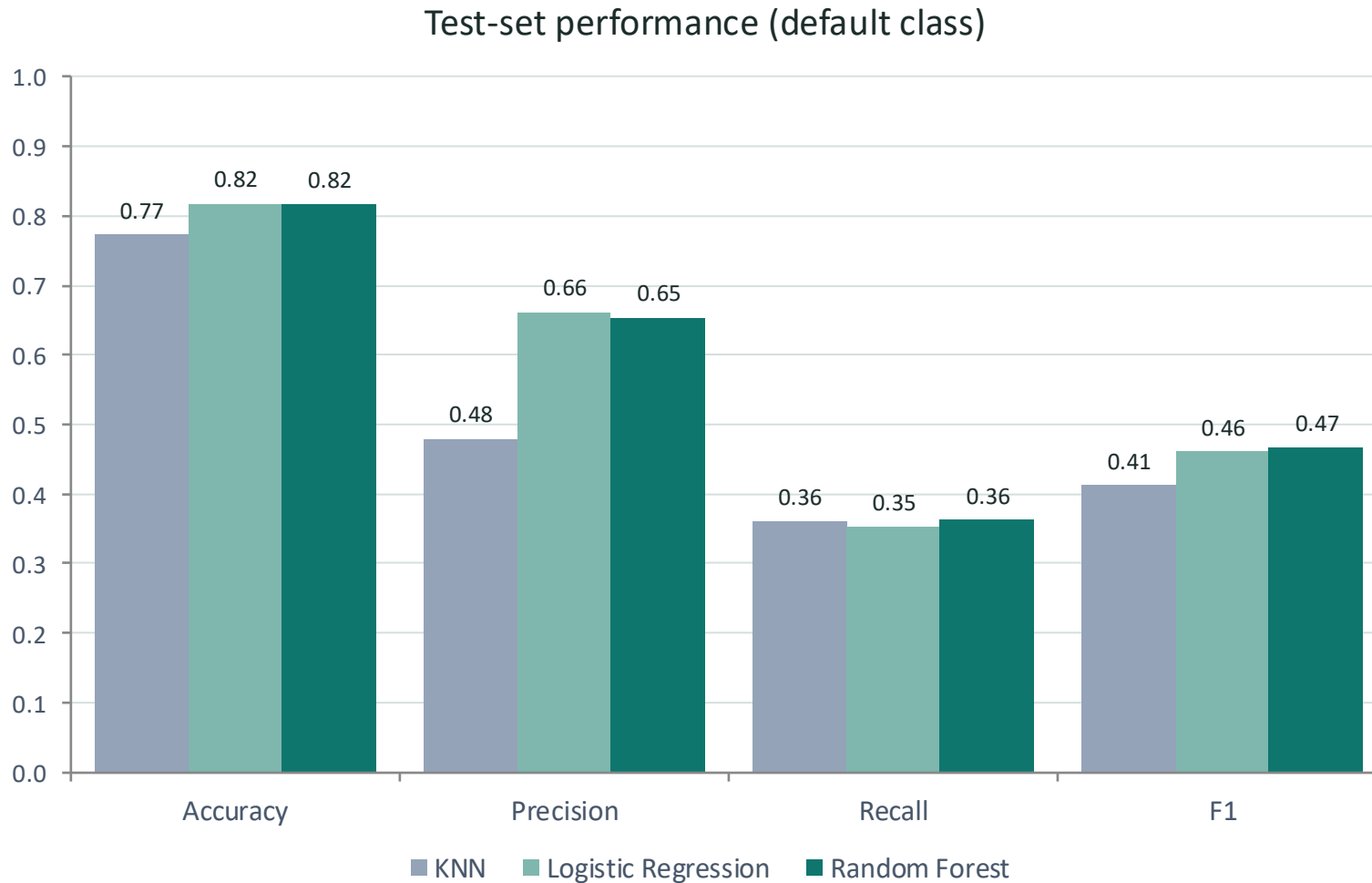
BEST PARAMETERS

n_estimators = 100
max_depth = None
min_samples_split = 5

0.373 ± 0.009

5-fold CV recall

Random Forest edges ahead on recall



Takeaways

- Random Forest: highest recall (0.362) and F1 (0.466).
- Logistic Regression: slightly higher accuracy and precision.
- Recall differences between the models are tiny, so interpret with care.

Error analysis: where the model misses

	Predicted: no default	Predicted: default
Actual: no default	4,417 true negatives	256 false positives
Actual: default	846 false negatives	481 true positives

Random Forest confusion matrix (test set, 6,000 clients)

846

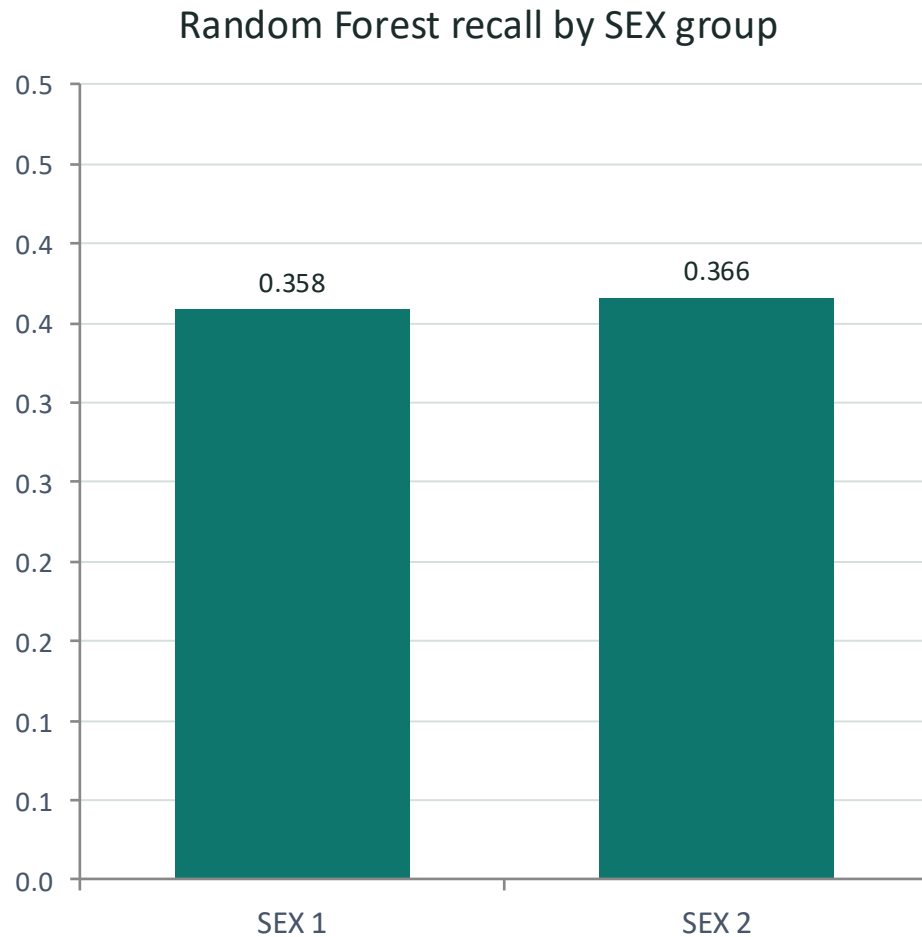
real defaulters predicted as safe

481

defaulters correctly caught

$\text{Recall} = 481 / (481 + 846) = 0.362$

Fairness and ethical limits



1

Subgroup recall is close

0.358 vs 0.366 between the two SEX groups, but it should be monitored in any real use.

2

Limited population

Taiwanese clients from 2005. It may not fit other countries or today's customers.

3

Sensitive variables

Sex, age, education and marital status raise fairness concerns.

4

Decision support only

A human reviews decisions. The model never auto-approves or denies credit.

Recommendation: support, not automation

Why Random Forest

- Highest recall: 0.362
- Highest F1: 0.466
- Returns a class and a default probability

Use with care

- Still misses about 2 in 3 defaulters
- Model gaps are small vs CV variation
- Not validated outside Taiwan

Next steps

- Try threshold tuning or class weights to lift recall
- Monitor subgroup performance
- Test on local data before wider use

Sources: UCI Machine Learning Repository, Default of Credit Card Clients (Yeh, 2009), doi.org/10.24432/C55S3H · scikit-learn · UN SDG 8 (sdgs.un.org)